

Tushika Garg

+91-7078323902 | gargtushikabsr@gmail.com | linkedin.com/in/tushika-garg | github.com/Tushika2024 | [LeetCode](#) | [Portfolio](#)

Computer Science undergraduate with strong fundamentals in Data Structures and Algorithms. Proficient in Python, C++, and SQL, with hands-on experience building ML applications and backend systems using agile development and RESTful API design. Completed a Software Developer internship working with production data.

EDUCATION

Thapar Institute of Engineering and Technology
B.E. in Computer Science and Business Systems; CGPA: 9.78/10.0

Patiala, Punjab

Aug 2024 – May 2028

Renaissance School
Senior Secondary Examination CBSE; Percentage: 95.6%

Bulandshahr, Uttar Pradesh

Apr 2022 – Mar 2023

WORK EXPERIENCE

Software Developer Intern

Jun 2026 – Jul 2026

Kanha Milk Testing Equipments Pvt. Ltd.

Bulandshahr, UP

- **Migrated 10,000+ Excel records** into MySQL via a Python ETL pipeline, resolving data format inconsistencies to ensure schema integrity and reduce query overhead.
- Developed a FastAPI-based backend service with automated analytics dashboards, applying REST API design and debugging practices to deliver data-driven visibility for stakeholders.

PROJECTS

CreditShield | Python, Sklearn, Optuna, SHAP, Streamlit | GitHub | Live

Apr 2026 – Jun 2026

- Engineered a **credit underwriting pipeline** on 30K+ records using Optuna-tuned XGBoost with **ROC-AUC 0.82**, reject inference, and **15+ behavioral risk features** to support data-driven credit decisioning.
- Designed a risk-segmented credit limit engine with SHAP explainability and configurable policy guardrails, delivering auditable, compliance-ready customer credit decisions via a cloud-deployed Streamlit application.

SentinelLabel | Python, Transformers, NLP, Sklearn, Cross-Encoders | GitHub

Feb 2026 – Apr 2026

- Developed a semantic label auditing framework across a 14K+ NLP corpus, flagging **268 mislabeled samples** with **82% precision**, achieving **3.6x improvement** over random sampling.
- Applied UMAP dimensionality reduction and HDBSCAN clustering to surface structural data inconsistencies, using modular, object-oriented design for testability and reuse.

Sleep Behavior Modeling for T2DM | Python, Sklearn, Pandas, NumPy | GitHub

Nov 2025 – Jan 2026

- Conducted EDA and statistical modeling on the AI-READI wearable health dataset, converting raw sensor data into structured, decision-ready features linked to Type-2 Diabetes risk.
- Achieved **ROC-AUC 0.75** and **F1-score 0.84** on T2DM risk classification; manuscript on Pattern Recognition in Wearable Sleep Data currently under conference review.

TECHNICAL SKILLS

Programming Languages: Python, SQL, C++, C, HTML, CSS

Machine Learning: Supervised & Unsupervised Learning, Model Training, Model Evaluation

Deep Learning & NLP: Neural Networks, Transformers, Text Classification, Embeddings

Libraries & Frameworks: ScikitLearn, Pandas, NumPy, Streamlit, FastAPI, Hugging Face

Tools: Git, GitHub, VS Code, MySQL

Coursework: Data Structures and Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, Software Engineering

ACHIEVEMENTS

- **Merit I Scholarship:** Awarded for achieving Rank 1 in Branch (First Year), Thapar Institute, 2024–25.
- **Flipkart Gridlock 2.0 (2026):** Top 1000 of 10,600+ teams, ML-based urban traffic congestion challenge.
- **GirlScript Summer of Code (2025):** Open Source Contributor for real-world projects via GitHub.
- **HackTU 6.0 (2025):** Led frontend development and ideation during a 24-hour rapid prototyping event.
- **Best Student of the Year:** Renaissance School, Session 2022–2023, for all-round excellence.
- **Competitive Programming:** Solved 300+ DSA problems on [LeetCode](#), improving code optimization and algorithmic efficiency.

LEADERSHIP & EXTRA-CURRICULAR

- **Institution of Engineers (IEI):** Former Technical Department Member.
- **Student Council:** Batch Representative, communicating student concerns to faculty and coordinating between peers and administration.
- **Design Thinking (Course Project):** Conducted empathy interviews with 20+ students in a 4-member Design Thinking team; defined problem statements around classroom engagement and communication barriers.